Victoria McTrusty

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Final year Computer Science Student at the University of Edinburgh, and previous Adobe intern (Summer 2024). Proficient in Python, Java, JavaScript, TypeScript, Scala, C, and assembly. Thrives in collaborative environments, passionate learner and enthusiastic about expanding skills in Software Development. Keen interests in Machine Learning, Ethical Computer Science, Computer Security & Natural Language Processing.

Education

University of Edinburgh - Masters of Informatics (MInf) **(September 2021 – June 2026)**
Predicted grade: 2:1

- 4th Year Dissertation: *'Ethical Computer Vision: Creating a video analysis pipeline for Face, Demographic, and Object Detection'*.
  - I set out to investigate whether ethical concerns commonly associated with facial analysis systems persist in other areas of computer vision - specifically, action recognition models.
  - Analysed the performance of a newly developed action recognition model and it's training dataset, drawing key conclusions about the racial makeup of the dataset. Making use of Python, DeepFace and RetinaFace.
  - Inspired discussions regarding the ethical benefits of race in action recognition datasets and whether other measures are better suited to capturing the vast array of human movement.
- 5th Year Dissertation: *'Regulating the Regulation: An Investigation into current legislation accessibility'*.
  - The current AI/Law landscape is difficult to navigate as large numbers of documents are being published every few days from industry and international institutions. Making use of Python and the HuggingFace model sentence_transformers.
  - I set out to create a Natural Language Processing tool to automatically parse these documents, to get an overview of things like frequency and variability and to help visualise these in readable formats. To even help support researchers by directing them to relevant sections of text.
- Core skills attained:
  - Developed a firewall, implemented several network protocols making use of Wireshark, implemented MitM attacks, successfully spoofed emails, carried out buffer overflow attacks, SQL injections, and exploited race conditions. Good understanding of memory corruption errors, SQL injection, race conditions, web attacks. Proficient in the following libraries and languages: Matplotlib, Pandas, DeepFace, RetinaFace, cv2, Spark, HuggingFace, Wireshark, Haskell.

George Heriot's School **(September 2013 – June 2021)**

- National 5: Computing (A), Math (A), English (A), Biology (A), Chemistry (A), French (A), Chinese (A)
- Highers: English (A), Math (A), Chemistry (B), Human Biology (A), Chinese (A), Computing (A), Statistics (Pass) – (pass/fail course)
- Advanced Highers: Chemistry (B), Biology (B)

Work Experience

Software Development Engineering Intern **(June 1st - August 31st 2024)**
Adobe

- I worked on the Admin Console - the website where Adobe's Enterprise customers manage their users licenses for their organisations.
- Created tooling to automate the identification of unused and duplicated React components.

- Migrated legacy components to a new internal component library, and these changes were deployed to production.
- I gained skills in front-end development (React, JavaScript, TypeScript), version control (Git), and scripting with Python.

Sales Assistant **(October 2022 - March 2023)**

Currys PC World

- I helped customers find products that best suit their needs and assist them through their sales journey with Currys. From this job I've garnered a lot of knowledge in computer specifications, knowledge in domestic appliances and improved my interpersonal skills.

Teacher **(April 2022 – December 2023)**

Code Ninjas

- I have been working at Code Ninjas in Morningside, Edinburgh. Where I have been a teacher to 4–14-year-olds explaining Java code and Python code. Other responsibilities include, classroom management, working with other colleagues and walking children through complicated Lego robotics activities. Over the summer I have also managed camps on topics such as Lego Robotics, Python, Roblox and movie making in Minecraft.

Technology **(July 8th– July 12th 2019)**

Mapix Technologies

- Work experience at a company that specialize in LiDAR (Light Detection and Ranging) technologies and their integration in mapping, autonomous vehicles and robotics.

Business Intern **(August 5th – August 9th 2019)**

PricewaterhouseCoopers (PWC)

- Gained insightful experience into the culture of PWC and learned about the possible career paths in professional consulting.

Experience/ Awards

- Arkwright Engineering Scholarship
- Combined Cadet Force 2016-2021, Cyber First Futures & Cyber First Advanced courses hosted by the National Cyber security centre
- A week in Intensive Care Unit

University Projects

Computer Security

- Introduced to network security and usable security, cryptography. Exposed to the knowledge of inevitable weakening of our standard encryption scheme protocols as they are based on intractable problems, due to the rise of quantum computing. Exploited OS security and carried out many attacks such as MitM attacks and buffer overflow attacks.

Communications and Networks

- Learnt about the different layers in the internet protocol suite. Implemented 3 different sliding window protocols, implemented a L4 firewall in OpenFlow, and carried out L4 Load Balancing in OpenFlow.

Applied Machine Learning

- Using data from the Spotify API, we created a song recommender and genre classifier using machine learning models such as a Logistic Regressor, Decision Tree Classifier, and Random Forest Classifier. We also made use of the lyric data and implemented natural language techniques to analyse semantic similarity.

Foundations of Natural Language Processing

- Built a character-level language model and implemented the Viterbi algorithm. Implemented a Hidden Markov Model in Python and trained it for part of speech tagging purposes using an emission and transition model.

References

Available upon request